COMP 1101 - Midterm 1

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1. Which of the following can not be a variable name?

a) count
b) toplam
c) sum
d) sum1
e) for

2. Given T as an integer, how do you check if T is between 0 and 11?

a) `if T < 0 || T > 11`
b) `if T < 0 && T > 11`
c) `if 0 < T < 11`
d) `if T > 0 || T < 11`
e) `if T > 0 && T < 11`

3. What is the value of integer variable x after the execution of the following line?

```
x = 1 / 2 + 1 / 2 + 1 / 4 + 1 / 4 + 1 / 2;
```

a) 0
b) 0.5
c) 1
d) 1.5
e) 2

4. Which of the following expressions result in value 2?

a) mod(44, 6)
b) mod(13, 3)
c) mod(20, 5)
d) mod(2, 1)
e) mod(19, 4)

5. Which of the following code pieces will print the value of the variable x to the screen?

a) `disp("value of x")`
b) `disp('x')`
c) `disp("x")`
d) `disp(value of x)`
e) `disp(x)`

6. Which statement reads an integer value from user?

a) `disp("Enter a new integer");`
b) `x = input("Enter a new integer);`
c) `input = ("Enter a new integer");`
d) `disp = ("Enter a new integer");`
e) `input("Enter a new integer);`

7. Suppose that x is a user given 3 digit integer (e.g. 197), which of the following prints its tens digit (e.g. 9)?

a) `mod(floor(197 * 10), 10);`
b) `mod(floor(197 / 100), 10);`
c) `mod(floor(197 / 10), 10);`
d) `floor(197 / 10);`
e) `mod(197, 10);`

8. Translate this statement into MATLAB: If the bmi is between 26 and 31, print "overweight".

a)
```
if bmi > 26 && bmi <31
   disp("overweight");
end
```

b)
```
if bmi >26 && bmi < 31;
   disp("overweight");
end
```

c)
```
if 26 > bmi < 31
   disp("overweight");
end
```

d)
```
if bmi > 26 and bmi < 31
   disp("overweight");
end
```

e)
```
if bmi > 26 and bmi < 31;
   disp("overweight");
end
```

9. Suppose x is 46, what is the output of the following code?

```
if x < 20
   disp("1");
else
   if x < 30
      disp("2");
```

```
    else
        if x < 50
            disp("3");
        else
            if x < 80
                disp("4");
            else
                disp("5");
            end
        end
    end
end
```

a) 1
b) 2
c) 3
d) 4
e) 5

10. What is the output of the following code, if the user enters 0?

```
choice = input("Enter a number: ");
switch choice
    case 0
        disp("Gunaydin");
    case 1
        disp("Good morning");
    case 2
        disp("Guten morgen");
    case 3
        disp("Bonjour");
    otherwise
        disp("Invalid input");
end
```

a) Good morning
b) GunaydinGood morning
c) Bonjour
d) Guten morgen
e) Gunaydin

11. Which of the following codes is the same as the following code?

```
if x == 2 || x == 4 || x == 9
    disp("Done");
else
    if x == 1 || x == 3
        disp("Not done");
    else
        disp("Invalid");
    end
end
```

a)

```
switch x
    case {2, 4}
        disp("Done");
    case {1, 3}
        disp("Not done");
end
```

b)

```
switch x
    case {2, 4, 9}
        disp("Done");
    case {1, 3}
        disp("Not done");
        end
```

c)

```
switch x
    case {2, 4, 9}
        disp("Done");
    case {1, 3}
        disp("Not done");
    case {5, 6}
        disp("Invalid")
end
```

d)

```
switch x
    case {2, 4, 9}
        disp("Done");
    case {1, 3}
        disp("Not done");
    otherwise
        disp("Invalid")
end
```

e) None of the above

12. Which of the following is the MATLAB counterpart for the following math expression?

$$x^2y + \frac{z}{t}$$

a) $x^2 * y + z/t$
b) $x * x * y + z/t$
c) $x^2 * y + zt$
d) $x^2y + \frac{z}{t}$
e) None of the above

13. What is the operator for checking equality of two variables in MATLAB?

a) $<>$
b) !=
c) ==
d) =
e) =!

14. What is the value of integer variable x after the execution of the following line?

```
x = 7 / 2 − 3 + 1 / 2;
```

a) 1
b) 5
c) 5.5
d) 0
e) 6

15. What is the value of the following MATLAB expression?

```
mod(19, 7)− 4
```

a) 1
b) 0
c) 12
d) 11.5
e) None of the above

16. Which of the following is the MATLAB counterpart for the following math expression?

$$\frac{x^2+2}{3y-5}$$

a) $(x*x+2)/(3y-5)$
b) $x*x+2/3*y-5$
c) $(x*x+2)/(3*y-5)$
d) $x^2+2/(3*y-5)$
e) $(x*x+2)/3y-5$

17. What is the output of the following program?

```
a = 4;
b = 8;
c = 12;
d = 8;
disp(a + b ∗ c + d − a ∗ c);
```

a) 160
b) 152
c) 240
d) 60
e) 108

18. Assume n is a MATLAB variable. Write a code to print to the screen positive if n is positive, zero if n is zero, and negative if n is negative

a)

```
if n > 0
   disp(”positive”);
else
   if n == 0
      disp(”zero”);
   else
      disp(”negative”);
   end
end
```

b)

```
if (n > 0)
   disp(”positive”);
else (n==0)
   disp(”zero”);
otherwise
   disp(”negative”);
```

c)

```
if n > 0
   disp(”positive”);
else
   if n = 0
      disp(”zero”);
   else
      disp(”negative”);
   end
end
```

d)

```
if n > 0
   disp(positive);
else
   if n == 0
      disp(zero);
   else
      disp(negative);
   end
end
```

e) None of the above

19. Translate this statement into MATLAB: If the bmi is less than 24 or larger than 30, print ”not normal”.

a)

```
if bmi < 24 && bmi > 30;
   disp(”not normal”);
end
```

b)

```
if bmi > 24 && bmi < 30
   disp(”not normal”);
end
```

c)

```
if 24 < bmi < 30
   disp(”not normal”);
end
```

d)

```
if bmi < 24 or bmi > 30
   disp(”not normal”);
end
```

e)

```
if bmi < 24 || bmi > 30;
    disp("not normal");
end
```

20. Suppose x is 51, what is the output of the following code?

```
if (x > 10)
    disp("1");
end
if (x > 40)
    disp("2");
end
if (x > 60)
    disp("3");
end
if (x > 80)
    disp("4");
end
```

a) 1
b) 1
   2
c) 3
d) 3
   4
e) 4

21. Which of the following codes reads his/her grade from the user and prints "Success" if the grade is above 50, "Fail" otherwise?

a)

```
grade = input("Enter the grade: ");
if grade >= 50
    disp("Fail");
else
    disp("Success");
end
```

b)

```
grade = input("Enter the grade: ");
if grade > 50
    disp("Fail");
else
    disp("Success");
end
```

c)

```
grade = input("Enter the grade: ");
if grade > 50
    disp("Success");
else
    disp("Fail");
end
```

d)

```
grade = input("Enter the grade: ");
if grade >= 50;
    disp("Success");
else
    disp("Fail");
end
```

e) None of the above

22. What are all possible outputs of the following code?

```
choice =input("Enter the choice number: ")
switch choice
    case 0
        disp("1");
    case 1
        disp("2");
    case 2
        disp("3");
    case 3
        disp("4");
    otherwise
        disp("0");
end
```

a) 1, 2, 3, 4
b) 1, 2, 3, 4, 0
c) 0, 40, 340, 2340, 12340
d) 1, 12, 123, 1234, 12340
e) 12340

23. Which of the following code prints (i) "Divisible by 2" if a given number N is divisible only by 2, and (i) "Divisible by 3" if a given number N is divisible only by 3, and (iii) "Divisible by 2 and 3" if a given number N is divisible by 2 and 3?

a)

```
if mod(N, 2) == 0
    disp("Divisible by 2");
else
    if mod(N, 3) == 0
        disp("Divisible by 2 and 3");
    else
        disp("Divisible by 3");
    end
end
```

b)

```
if mod(N, 2) == 0
    disp("Divisible by 2");
else
    if mod(N, 3) == 0
        disp("Divisible by 3");
    else
        disp("Divisible by 2 and 3");
```

```
    end
end
```

c)

```
if mod(N, 2) == 0
    if mod(N, 3) == 0
        disp("Divisible by 2 and 3");
    else
        disp("Divisible by 2");
    end
else
    disp("Divisible by 3");
end
```

d)

```
if mod(N, 2) == 0
    if mod(N, 3) == 0
        disp("Divisible by 2 and 3");
    else
        disp("Divisible by 2");
    end
else
    if mod(N, 3) == 0
        disp("Divisible by 3");
    end
end
```

e)

```
if mod(N, 2) == 0
    disp("Divisible by 2");
else
    if mod(N, 3) == 0
        disp("Divisible by 2 and 3");
    else
        disp("Divisible by 3");
    end
end
```

24. Which of the following code prints (i) "Divisible by 2" if a number N less than 10 is divisible only by 2, and (i) "Divisible by 3" if a a number N less than 10 is divisible only by 3, and (iii) "Divisible by 2 and 3" if a a number N less than 10 is divisible by 2 and 3?

a)

```
switch N
    case {2, 4}
        disp("Divisible by 2");
    case {3, 9}
        disp("Divisible by 3");
    case 6
        disp("Divisible by 2 and 3");
end
```

b)

```
switch N
    case {2, 4, 8}
        disp("Divisible by 2");
    case {3, 9}
        disp("Divisible by 3");
    otherwise
        disp("Divisible by 2 and 3");
end
```

c)

```
switch N
    case {2, 4, 8}
        disp("Divisible by 2");
    case {3, 9}
        disp("Divisible by 3");
    case 6
        disp("Divisible by 2 and 3");
end
```

d)

```
switch N
    case {2, 4, 8, 10}
        disp("Divisible by 2");
    case {3, 9}
        disp("Divisible by 3");
    case 6
        disp("Divisible by 2 and 3");
end
```

e) None of the above

25. Which of the following codes is the same as the following code?

```
if x == 2 || x == 4
    disp("Even");
else
    if x == 1 || x == 3
        disp("Odd");
    else
        disp("Invalid");
    end
end
```

a)

```
switch x
    case {2, 4}
        disp("Even");
    case {1, 3}
        disp("Odd");
    otherwise
        disp("Invalid");
end
```

b)

```
switch x
    case {2, 4, 6}
        disp("Even");
    case {1, 3, 5}
        disp("Odd");
end
```

c)

```
switch x
    case {2, 4}
        disp("Even");
    case {1, 3}
        disp("Odd");
end
```

d)

```
switch x
    case {2, 4, 6}
        disp("Even");
    case {1, 3, 5}
        disp("Odd");
    otherwise
        disp(" Invalid ");
end
```

e)

```
switch x
    case {2, 4}
        disp("Odd");
    case {1, 3}
        disp("Even");
    otherwise
        disp(" Invalid ");
end
```